

onero Actuator User Guide

1. Safety and Precautions

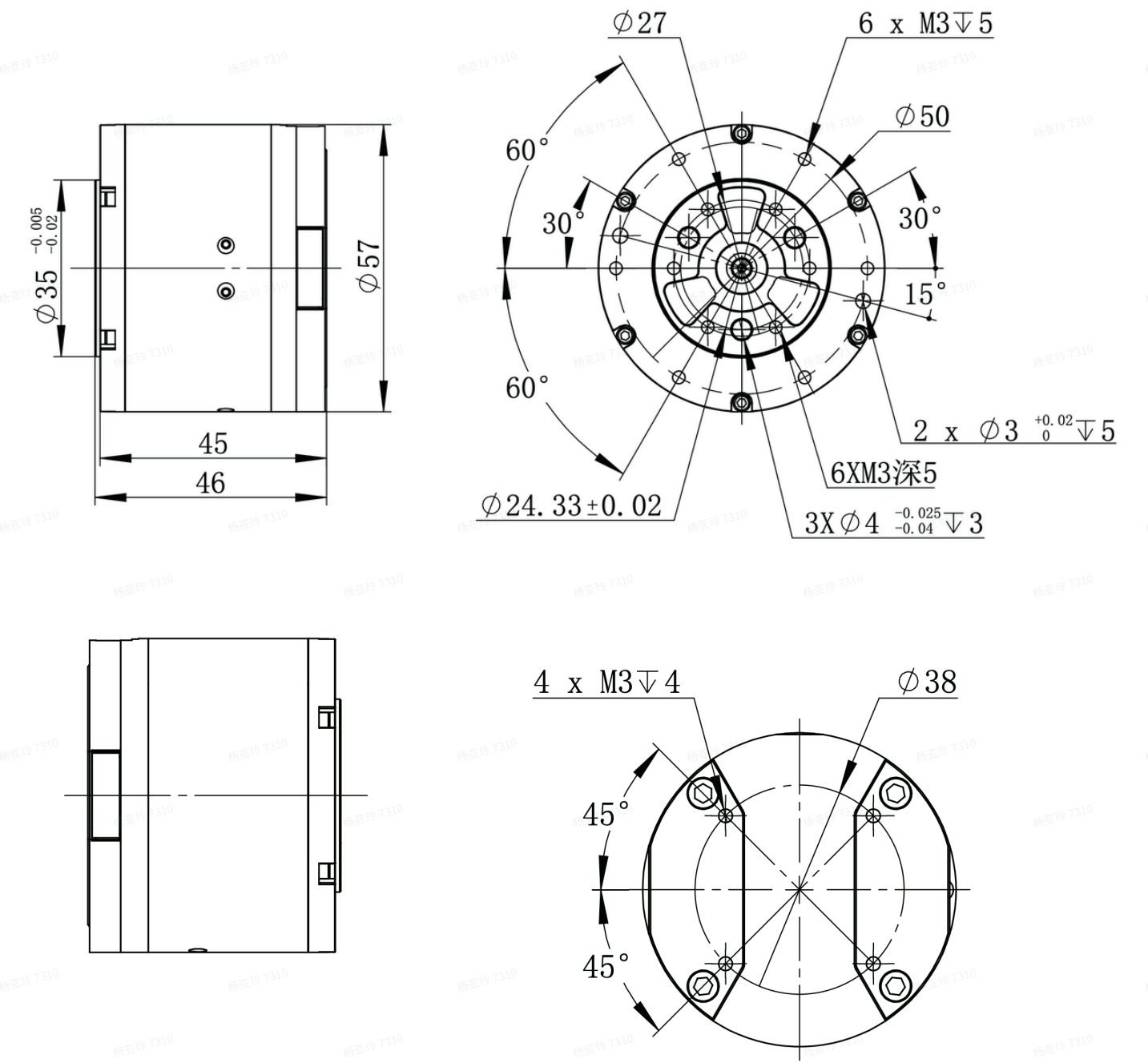
- Operate the actuator strictly within the specified environmental conditions and the maximum allowable winding temperature range. Exceeding these limits will cause permanent and irreversible damage to the product.
- Prevent foreign objects from entering the rotor assembly, as this will cause abnormal operation of the actuator.
- Before use, inspect all components for integrity. Discontinue use immediately if any parts are missing, aged, or damaged.
- Ensure all electrical connections are correct and that the actuator is properly and securely mounted.
- Do not touch the rotating parts or electronic components of the actuator during operation to prevent accidents.
- The actuator may become hot during high-torque output. Take care to avoid burns.
- Do not disassemble the actuator. Unauthorized disassembly will adversely affect its control precision and may lead to operational failure.

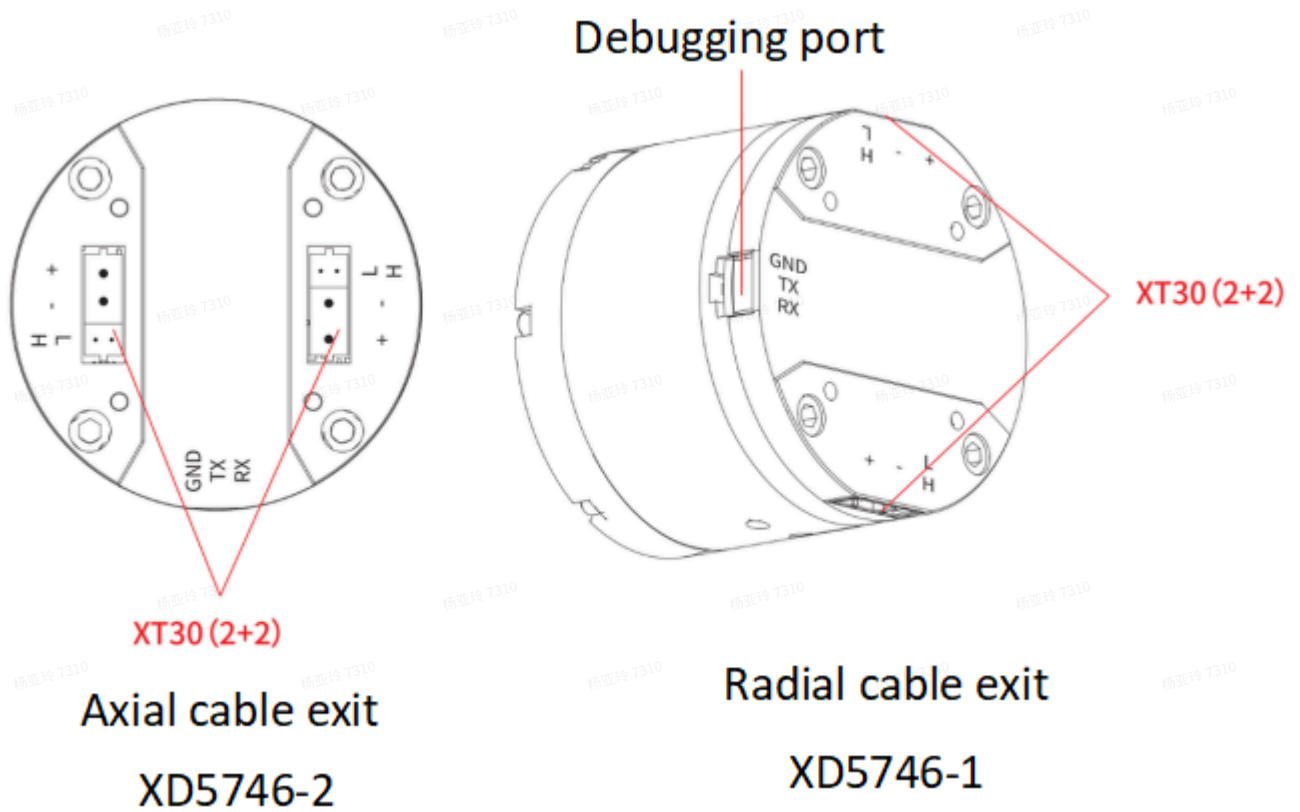
2. Product Features

- All-in-one joint module with a compact structure
- Compact size, superior torque output
- High-precision 21-bit encoder
- Dual encoder system: provides absolute single-turn positioning of the output shaft; position is maintained during power loss
- Comprehensive protection: over-temperature, over-voltage, over-current, under-voltage, and timeout
- High-speed CAN 2.0 communication interfaces
- Multiple control modes: position, velocity, torque, and MIT hybrid control mode
- Wide power supply range: supports 20V to 55V input
- Flexible wiring: side-exit and rear-exit options for easy mechanical integration

3. XD5746-1/2

3.1 Dimensions





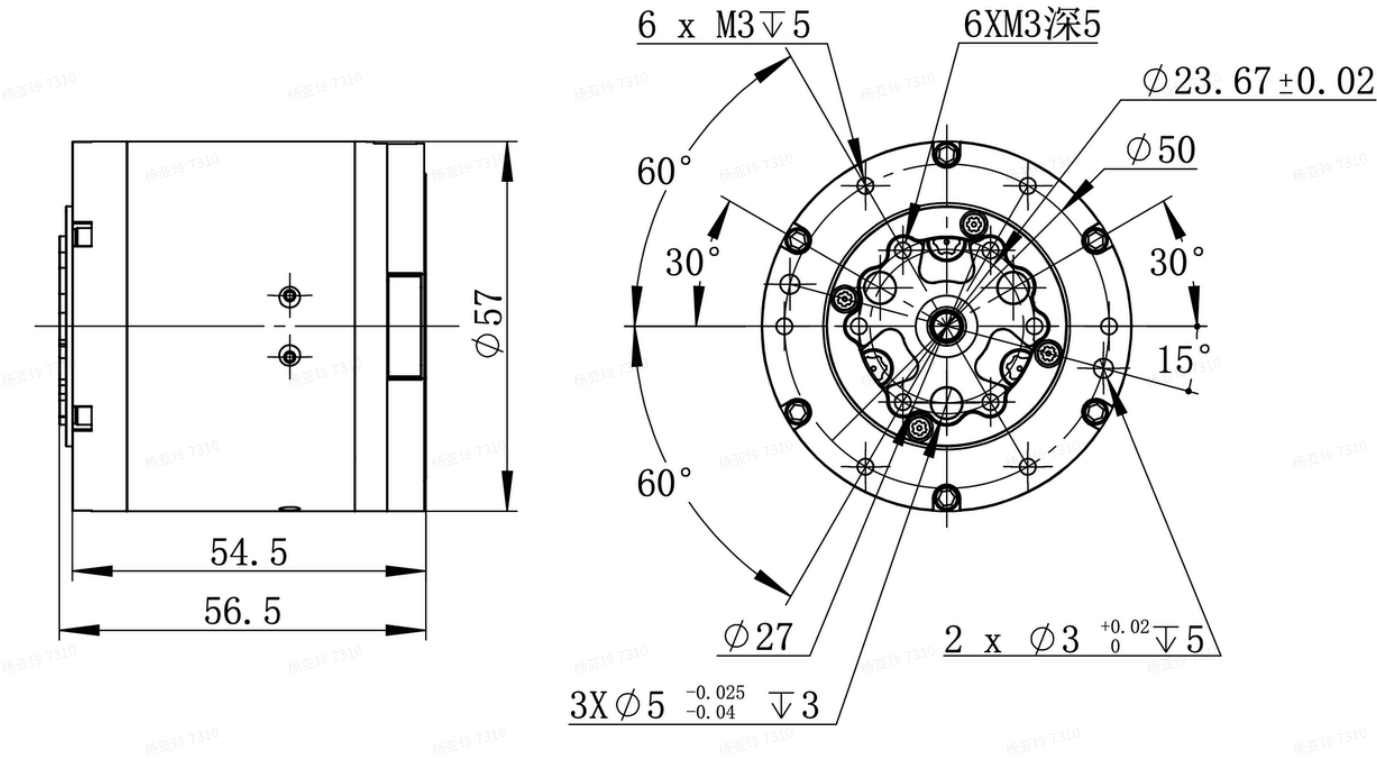
3.2 Technical Specifications

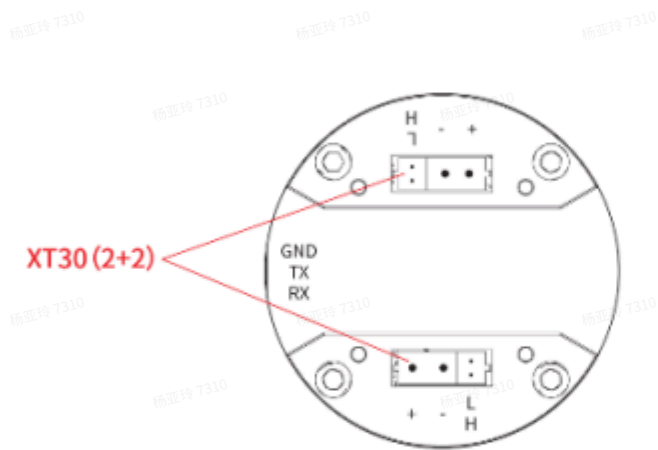
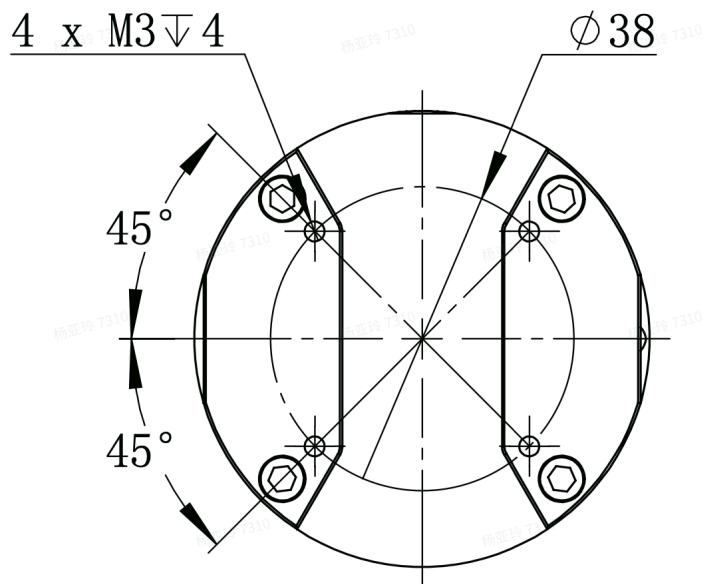
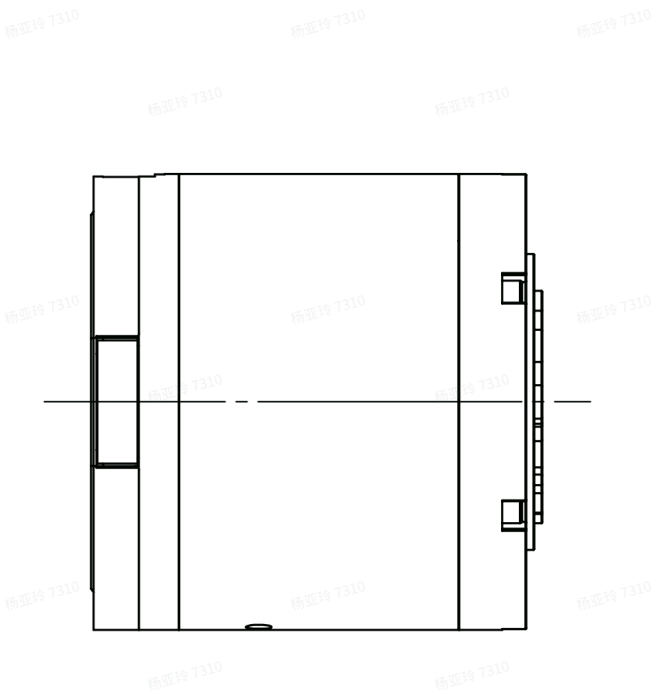
Category	Item	Value
Performance Parameters	Rated Voltage	48V
	Rated Torque	3N.m
	Rated Rotational Speed	120rpm
	Rated Current	3.7A
	Operating Voltage Range	20V~55V
	Peak Torque	7N.m
	Maximum No-Load Rotational Speed	200rpm
	Peak Current	7.4A
	Number of Pole Pairs	14
	Gear Ratio	10:1
Mechanical Parameters	Rotational Backlash	10 arcminutes
	Diameter	57mm

	Height	46mm
	Total Weight	320g
	Operating Temperature	5-50℃
Interface Parameters	Operating Humidity	RH 35% to 80%, non-condensing
	Electrical Connector	CAN@1Mbps
	Communication Terminal	XT30(2+2)

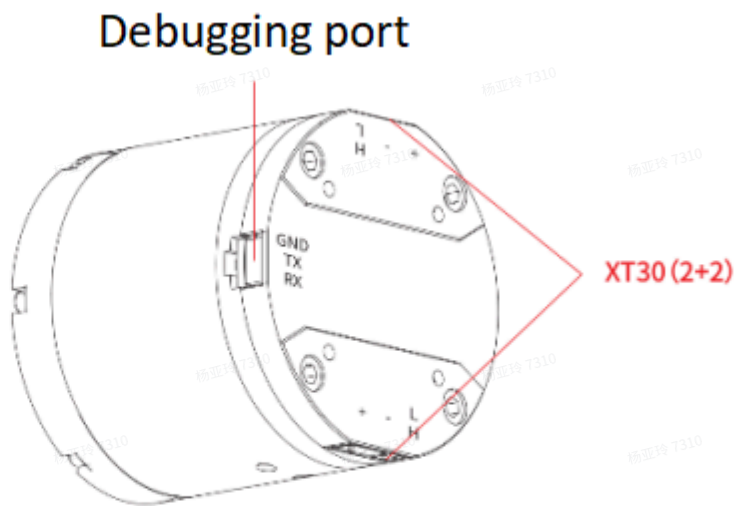
4. XD5757-1/2

4.1 Dimensions





Axial cable exit
XD5757-2



Radial cable exit
XD5757-1

4.2 Technical Specifications

Category	Item	Value
Performance Parameters	Rated Voltage	48V
	Rated Torque	15N.m
	Rated Rotational Speed	25rpm
	Rated Current	3.7A
	Operating Voltage Range	20V~55V
	Peak Torque	30N.m
	Maximum No-Load Rotational Speed	200rpm
	Peak Current	7.4A
	Number of Pole Pairs	14
	Gear Ratio	48.19:1
	Rotational Backlash	20 arcminutes
Mechanical Parameters	Diameter	57mm
	Height	56.5mm
	Total Weight	390g
	Operating Temperature	5-50°C
	Operating Humidity	RH 35% to 80%, non-condensing
Interface Parameters	Electrical Connector	CAN@1Mbps
	Communication Terminal	XT30(2+2)

5. Communication Protocol

Click the link below to view the "Actuator Communication Protocol".

- English: [Communication Protocol for the onero Actuator.pdf](#)
- Chinese: [onero Actuarot通信协议.pdf](#)

6. Debugging Tool

Click the link below to view the "Debug Tool Quick Guide".

- English: [Debugging Tool Quick Guide.pdf](#)
- 中文版: [调试工具使用说明.pdf](#)
- Debug software download: [motor_debug_tool_v1.0.2.zip](#)

7. Mechanical Drawings

- XD5746-1
 - 2D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5746-1_2d.pdf
 - 3D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5746-1_3d.stp
- XD5746-2
 - 2D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5746-2_2d.pdf
 - 3D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5746-2_3d.stp
- XD5757-1
 - 2D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5757-1_2d.pdf
 - 3D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5757-1_3d.stp
- XD5757-2
 - 2D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5757-2_2d.pdf

- 3D Drawing:
https://github.com/OpenWonderLabs/RobotDoc/blob/main/cad/actuator/XD5757-2_3d.stp

8. Related Documents

- Actuator User Guide (中文版): [关节说明书.pdf](#)
- A1 Robotic Arm Product Manual: [A1_Product_Manual.pdf](#)
- A1 Robotic Arm User Guide: [A1_User_Guide.pdf](#)
- A1 机械臂产品说明书 (中文): [A1_产品说明书.pdf](#)
- A1 机械臂快速入门手册 (中文): [A1_快速入门手册.pdf](#)

9. SDK & API

- C/C++, Python SDK: <https://github.com/OpenWonderLabs/ArmApi>
- ROS2: <https://github.com/OpenWonderLabs/ArmRos>